

Summer Math Adventure

W5
D1

Week 5 · Day 1 | Angles: Types, Measuring & Drawing

Name: _____

Date: _____

GEOMETRY: ANGLES

Angle Types: Acute $< 90^\circ$ | Right $= 90^\circ$ | Obtuse: $90^\circ - 180^\circ$ | Straight $= 180^\circ$ | Reflex: $180^\circ - 360^\circ$

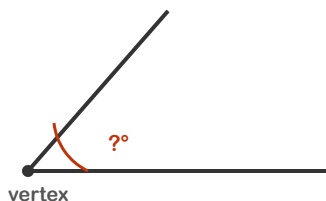
Using a protractor: Place center hole on vertex. Line up one ray on 0° . Read the scale where the other ray crosses.

Remember: Acute $\rightarrow$ use the SMALLER number; Obtuse $\rightarrow$ use the LARGER number on the protractor.

Part A – Name That Angle

Identify each angle type and estimate the measure. Then use your protractor to measure exactly.

Angle 1

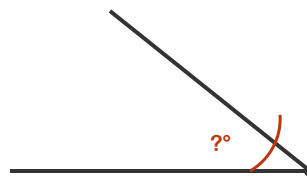


Type:

Measured:

°

Angle 2

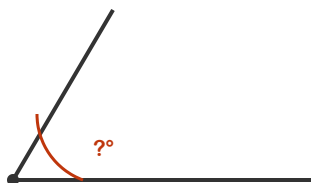


Type:

Measured:

°

Angle 3

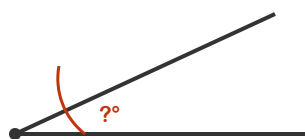


Type:

Measured:

°

Angle 4



Type:

Measured:

°

Part B – Draw the Angle

Use your protractor and the starting ray given. Draw the second ray to form the angle.

Draw a 35° angle.

Draw a 110° angle.

Draw a 75° angle.

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Week 5 · Day 2 | Lines, Triangles & Quadrilaterals

Part C – Angle Arithmetic

Name: _____

Date: _____

1. Two angles together make a right angle (90°). One angle is 37° . What is the other?

2. Two angles together make a straight line (180°). One is 124° . What is the other?

Line types: Line (infinite both ways), Ray (one endpoint), Line Segment (two endpoints)

Parallel lines never meet. Perpendicular lines meet at 90° .

Triangle angles always add to 180° . Quadrilateral angles always add to 360° .

Triangle types: Equilateral (3 equal), Isosceles (2 equal), Scalene (none equal), Right (one 90° angle)

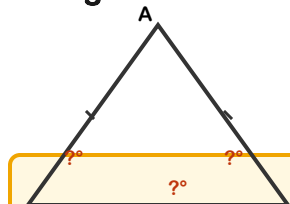
fourth?

Angles: _____

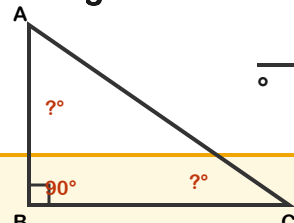
Part A – Classify the Triangle

Use your protractor to measure the angles. Then classify by angles **AND** by sides.

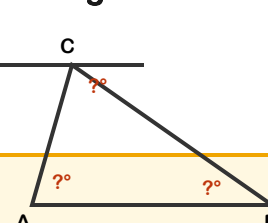
Triangle 1



Triangle 2



Triangle 3



★ **Challenge – Clock Angles**

Angle A:

A clock shows 3:00. The minute hand points at 12 and the hour hand at 3. What angle is formed?

At 6:00, what angle? At 2:00, what angle? (Hint: $360^\circ \div 12 \text{ hours} = 30^\circ$ per hour mark)

Angle B:

Angle B: 90°

Angle B:

Angle C:

Angle C:

Angle C:

2:00 →

Type (sides):

Type (sides):

Type (sides):

Type (angles):

Type (angles):

Type (angles):

Part B – Find the Missing Angle

1. Triangle: angles are 48° , 75° , and $?$
 $?$ = _____

2. Triangle: angles are 90° , 36° , and $?$
 $?$ = _____

3. Quadrilateral: angles are 85° , 110° , 95° , and $?$
 $?$ = _____

4. Parallelogram: one angle is 68° . Find all 4 angles.

(Hint: opposite angles are equal; adjacent angles add to 180°)

Angles: _____

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Write the name of the quadrilateral that matches each description.

Week 5 · Day 3 | Area & Perimeter of 2D Shapes

Name:	Description	Name
Date:	4 equal sides, 4 right angles	
	Opposite sides equal and parallel, 4 right angles	

AREA & PERIMETER

Perimeter = total distance around the outside (add all sides). Units: cm, m, ft, in.

Area = space inside a shape. Units: cm², m², ft², in².

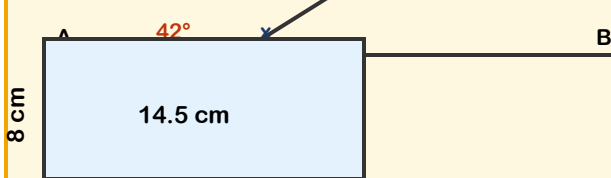
• Rectangle: $A = \text{length} \times \text{width}$ $P = 2(l + w)$

• Triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$

• Parallelogram: $A = \text{base} \times \text{height}$ (height is perpendicular distance, NOT the slanted side!)

In the figure below, line AB is straight. Find angles x, y, and z. Show your reasoning.

Shape 1 – Rectangle



$x =$
Perimeter =

$y =$
cm

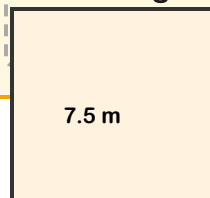
Area =

$z =$

cm²

Shape 3 – Square

Reasoning:



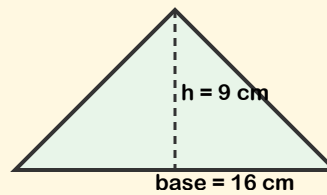
Perimeter =

m

Area =

m²

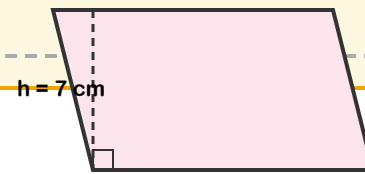
Shape 2 – Triangle



Area =

cm²

Shape 4 – Parallelogram



Area =

cm²

Part B – Composite Shapes

Break the shape into rectangles or triangles, find each area, then add.

Find the area of the L-shaped figure:

10 ft

8 ft

4 ft

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Week 5 · Day 4

Volume of Rectangular Prisms & Coordinate Plane

Name: _____

Date: _____

VOLUME & COORDINATES

Break into rectangles. Label and solve:

Volume = length × width × height. Units: cm³, m³, ft³, in³.
 Think of it as: (base area) × height, or count cubic unit cubes.
Coordinate plane: (x, y) → x = right/left, y = up/down. Origin = (0,0).
 Quadrant I: (+, +) II: (−, +) III: (−, −) IV: (+, −)

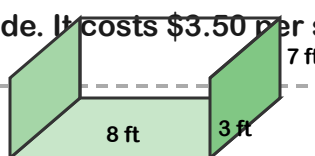
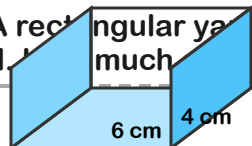
Part A – Volume of Rectangular Prisms

Prism 1

Part C – Word Problems

Prism 2

5. A rectangular yard is 24.5 feet long and 18 feet wide. It costs \$3.50 per square foot to install sod. How much will it cost to sod the entire yard?



Area:
 $V = l \times w \times h = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} =$

$V =$

ft^2 Cost: \$ _____
 cm^3

_____ ft^3

Word Problem: A fish tank is 24 inches long, 12 inches wide, and 18 inches tall. What is its volume?

6. A triangular garden has base 12 m and height 7.5 m. A fence costs \$18 per meter. The three sides measure 12 m, 9 m, and 10.5 m. How much does the fence cost?

$V =$

Perimeter:

_____ in^3

m Fence cost: \$ _____

Find the missing dimension: A box has volume 360 cm³. Its base is 12 cm × 6 cm. What is the height?

★ Challenge – Same Area, Different Perimeter

Height = _____
 Draw TWO different rectangles that each have an area of 24 cm². Find the perimeter of each. Which has the smaller perimeter?

Rectangle 1: _____ × _____ = 24 P = _____

Rectangle 2: _____ × _____ = 24 P = _____

Smaller perimeter: Rectangle

Part B – Coordinate Plane

Use the grid below. Label the x-axis and y-axis. Plot each point and label it.

Summer Math Adventure

Week 5 · Day 5 | FULL CUMULATIVE REVIEW – All 5 Weeks

Name: _____

Date: _____

W5
D5

CUMULATIVE REVIEW

You've covered: Place value · Rounding · Add/Subtract big numbers · Multiplication · Division · Factors/Multiples · Order of Operations · Fractions · Mixed Numbers · Decimals · Angles · Area/Perimeter · Volume · Coordinates

This is your chance to show how much you remember!

Plot and label these points:
A (3, 2) B (-1, 4) C (-3, -2)
D (2, -3) E (0, 3) F (-4, 0)

Questions:

1. Which quadrant is A in?

2. Which quadrant is C in?

1. Write $40,000,000 + 300,000 + 5,000 + 70 + 2$ in standard form:

4. Plot point G so that ABGD forms a rectangle. What are G's coordinates?

3. $8,040,000 - 3,857,621 =$

4. $648 \times 75 =$

★ Challenge – Coordinate Geometry

Points P(1,1), Q(5,1), R(5,4) form a triangle. Find the area of this triangle using coordinates. (Hint: it's a right triangle — use the legs as base and height.)

5. $7,344 \div 48 =$

6. $\text{GCF}(36, 54) =$

Area =

$\text{LCM}(9, 15) =$

square units

7. $4 + 3 \times (8 - 2)^2 \div 9 =$

8. $\frac{5}{6} + \frac{3}{8} =$

9. $5\frac{1}{4} - 2\frac{5}{6} =$

10. $\frac{3}{4} \times \frac{8}{9} =$

11. $7.4 - 2.863 =$

12. $3.6 \times 0.25 =$

Word Problems

13. A swimming pool is 50 m long, 25 m wide, and 2.4 m deep. What is its volume? If water costs \$0.04 per cubic meter, how much does it cost to fill the pool?

Topics: Angles · Shapes · Area/Perimeter · Volume · Coordinates + Cumulative

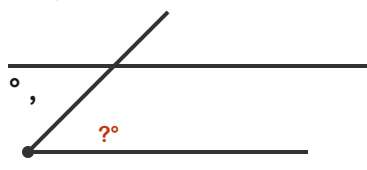
Name: _____

Date: _____

14. A triangle has angles in the ratio 1:2:3. What are the three angles?

Section 1 – Angles (5 pts)

1. Measure this angle with your protractor:



Measure: _____

Type: _____

2. Two angles form a straight line. One is 67° . Find the other.

★ Final Challenge – The Math Maze

Start with the number 1,000. Follow each instruction and write the result after each step.

3. Triangle angles: 52° , 78° , $?$.
1. Add 3,847: _____

2. Multiply by $\frac{1}{2}$: _____

3. Subtract 248.5: _____

5. (1 pt) A quadrilateral has angles 88° , 112° , 95° . Find the fourth angle.

4. Divide by 0.5: _____

5. Find 30% of your result ($30\% = 0.3$): _____

6. Add the LCM of 8 and 12: _____

Section 2 – Area, Perimeter & Volume (9 pts)

6. Rectangle: base = 15 m, height = 7 m.

Area: _____

Check each step:

Perimeter: _____

7. Triangle: base = 10 m, height = 6.4 m

Area: _____

m²

8. (3 pts) A room is 15 ft long, 12 ft wide, and 9 ft tall. Find: (a) the area of the floor, (b) the volume of the room, (c) how much it costs to carpet the floor at $\$4.50/\text{ft}^2$.

Floor area: _____

ft³ Carpet cost: \$

9. (3 pts) A rectangular prism has volume 504 in³. The base is 12 in × 7 in. Find the height.

Height =

in

Section 3 – Cumulative Final (6 pts)

10. $487 \times 63 =$

11. $5\frac{1}{3} + 2\frac{3}{4} =$

12. (4 pts) A hiker walks 4.75 km on Day 1 and $\frac{3}{4}$ of that on Day 2. On Day 3, she walks twice as far as Day 2. What is the total distance for all 3 days? Give your answer as a decimal.

Total:

km